

Impact Model - ET Co-Op AI Household Money Mentor

Goal

Estimate business impact in three dimensions:

1. Time saved in planning and execution.
2. Cost reduced in delivery operations.
3. Financial value recovered from tax leakage (and optional monetization).

Assumptions (Base Case)

Volume and operating assumptions

1. Households served per year: 500
2. Fully loaded analyst/operations cost: INR 900 per hour

Manual workflow effort per household (before ET Co-Op)

1. Data collection and sanity checks: 35 minutes
2. Old/new regime comparison and deduction review: 40 minutes
3. Additional tax-leakage and allocation verification: 30 minutes
4. Recommendation write-up: 20 minutes
5. Drafting two HR emails: 20 minutes
6. Final quality check and corrections: 15 minutes
7. Total manual time: 160 minutes

ET Co-Op workflow effort per household (after ET Co-Op)

1. Input capture and validation in UI: 25 minutes
2. Optimization run and review: 10 minutes
3. AI narrative and HR draft generation: 7 minutes
4. Final review and action execution: 8 minutes
5. Total assisted time: 50 minutes

Financial recovery assumptions

1. Average tax leakage identified per household: INR 60,000
2. First-year realization rate (implemented recommendations): 35%

Optional monetization assumptions

1. Conversion to premium support tier: 10% of households
2. Annual premium support fee: INR 2,000 per converted household

Back-of-Envelope Math

1) Time saved

Time saved per household:

$$160 - 50 = 110 \text{ minutes} = 1.83 \text{ hours}$$

Annual time saved:

$$500 \times 1.83 = 915 \text{ hours/year}$$

2) Cost reduced

Annual delivery cost reduced:

$$915 \times 900 = 823,500 \text{ INR/year}$$

Approximate: INR 8.24 lakh/year

3) Financial value recovered

Annual tax value recovered for households:

$$500 \times 60,000 \times 0.35 = 10,500,000 \text{ INR/year}$$

Approximate: INR 1.05 crore/year

4) Optional revenue recovered (monetized service layer)

$$500 \times 0.10 \times 2,000 = 100,000 \text{ INR/year}$$

Approximate: INR 1.0 lakh/year

Scenario Sensitivity

Scenario	Households/year	Time saved per household	Cost reduced/year	Leakage/household	Realization	reco
Low	300	90 min (1.5 h)	INR 405,000	INR 45,000	25%	INI
Base	500	110 min (1.83 h)	INR 823,500	INR 60,000	35%	INR
High	1,000	120 min (2.0 h)	INR 1,800,000	INR 75,000	40%	INR

Interpretation

- 1. The strongest impact is value recovery from reducing tax leakage for households.
- 2. Operational savings are material and predictable even without external AI/email keys.
- 3. Full provider integration (OpenAI + Resend) improves speed and consistency, lifting both throughput and conversion potential.

Limitations and Notes

- 1. This is a directional model for judging and planning, not audited finance.
- 2. Realization rates depend on user follow-through and payroll timelines.
- 3. For stricter ROI tracking, collect cohort-level data for submitted declarations, accepted deductions, and actual tax paid over time.